

Saransh (सारांश)

Multimodal Healthcare Triage Assistant for Government and Institutional Health Facilities

Human-in-the-loop, non-diagnostic triage support transforming chaotic multilingual waiting rooms into structured, prioritised clinical lanes.

Team: CODEX | Team ID: BH26PS07T060 | Educational prototype for triage support only. Synthetic data only.

Rules Decide, LLM Summarises

Urgency flags come from transparent clinical rules; AI never diagnoses.

Vernacular + Voice First

Odia, Hindi & regional speech in; standardized clinical English note out.

Missing-Info Engine

Detects absent clinical history and prompts frontline staff with follow-ups.

One Tool, Many Settings

PHC, CHC, campus fever bay, maternal check-in, and rural health camps.

Ideation

A safety-first assistant that turns messy symptoms, voice and reports into one structured, prioritised note for a qualified reviewer.

- **Rules decide, LLM summarises**

Urgency flags come from transparent deterministic rules; AI never diagnoses or prescribes.

- **Vernacular + voice first**

Spoken Hindi, Odia and other regional languages in; standardized clinical English note out.

- **Missing-info engine**

Actively finds what is absent from intake and prompts the frontline worker with targeted follow-ups.

- **One tool, many settings**

Scalable profiles: General OPD queue, factory clinic, campus fever bay, maternal care, and mobile health camps.

⚠️ Why Existing Models Fall Short:

- Symptom checkers: Diagnostic-first, English-only, high hallucination risk, consumer-facing rather than clinical staff.
- Telemedicine platforms: Facilitates video consults, but lacks structured frontline intake and emergency queue sorting.
- Point-solution Imaging AI: Narrow focus on radiology images; ignores multimodal vernacular intake and referral continuity.
- HMIS / Heavy EHRs: Clunky manual typing, complex forms, negligible regional voice NLP, fails on low-bandwidth rural devices.

🔄 Our Strategic Gap & Breakthrough:

A lightweight, reviewer-facing clinical triage layer placed directly in front of existing government hospital workflows — bridging vernacular patients and busy medical officers without altering existing backend systems.

Problem Relevance

Directly matches the government brief across PHCs, CHCs, health camps, industrial clinics and campus centres.

1. Overcrowded OPDs & Camps

200–400 patients arrive during crowded morning hours; doctors get under 90 seconds per patient with zero structured intake.

2. Language Diversity

Patients speak regional languages (Odia, Hindi dialects), while medical documentation and clinical notes are standardized in English.

3. Paper Reports & Photos

Reports are crumpled paper or camera photos; critical numeric parameters, previous trends, and timelines get lost in the rush.

4. Specialist Scarcity

Rural PHCs and camps lack specialists; handoff transfer notes are weak or scribbled on scrap paper without baseline vitals.

5. Unprioritized First-Come Queues

Urgent emergencies (silent hypoxia SpO2 < 90%, crushing chest pain) wait in the exact same line as mild routine check-ins.

6. Uneven Digital Maturity

Rural facilities face low bandwidth, shared mobile devices, and lack formal EHR hardware at frontline stations.

 **The Essential Gap: Structured, language-ready, prioritised information reaching the right healthcare professional in time.**

Solution

A 4-step pipeline turning multimodal frontline inputs into structured, reviewer-facing priority intelligence.

1. Collect

Patient or worker enters symptoms by text or voice in English, Hindi, or Odia.

- Audio voice recording
- 2D Anatomical Body Map
- ABHA ID scan & consent

2. Extract

Multimodal OCR reads lab reports and prescriptions; values and dates become a timeline.

- CBC, ECG, report OCR
- Numeric value parsing
- Chronological timeline

3. Flag

Rules-based risk flags and category tags set an urgency signal. No diagnosis.

- Hard SpO2 < 90% floor
- MAX(Rule, AI) logic
- Instant Red alert tone

4. Review

Nurse or doctor sees a structured note, follow-up questions and queue order, then decides.

- 3-Lane doctor queue
- Clinician override log
- 1-Click referral slip



Standardized Clinical Outputs & Multi-Setting Scenarios:

- Core Outputs: One-page triage note + prioritised clinical queue (● RED, ● YELLOW, ● GREEN) + referral draft, always labelled advisory and reviewer-facing.
- Supported Scenarios: Primary OPD queue, industrial-estate screening, campus fever bay, maternal follow-up, chronic check-in, health camps, and inter-facility referral notes.

Impact

Empowering patients, frontline workers, and institutional facilities with measurable pilot metrics.

Patients

- Speak in their own mother tongue (Odia/Hindi).
- Urgent and critical cases reach clinicians sooner.
- Clear, tamper-evident referral handoff documentation.
- Zero cost and zero complex device requirements.

Health Workers

- Less typing and repetitive documentation overhead.
- Guided follow-up questions to uncover missing history.
- Consistent, standardized clinical handoff notes.
- Easy 2D body map for non-literate patients.

Facilities

- Better queue order and structured emergency lanes.
- Auditable clinician override records and audit trails.
- Stronger, verified referral links to higher tertiary centres.
- Real-time emergency bay bed telemetry tracking.

Pilot Success Metrics (To be measured objectively, not assumed):

- 🕒 Time to Reviewer Attention: Time from patient arrival to reviewer attention for flagged urgent cases (Target: <30 seconds).
- 📄 Report OCR Accuracy: Share of paper lab report values correctly extracted, verified against staff audits (Target: >95%).
- 🩺 Reviewer Agreement Rate: Reviewer edits per note and degree of agreement with deterministic urgency flags (Target: >90%).
- ⌚ Frontline Staff Time Saved: Staff-reported time saved per patient intake (Reduced from 8–10 minutes down to <2 minutes).

Technical Depth

Full-stack architecture engineered with transparent deterministic safety guardrails.

Frontend Layer

React 18 PWA, Vite, Tailwind CSS, Web Speech API, offline-first IndexedDB, responsive mobile & desktop view.

Backend Engine

FastAPI (Python 3.11+), Pydantic v2 strict schemas across all clinical domains, RESTful JSON API.

Speech & Translation

Sarvam AI (Saaras:v3 ASR, Bulbul:v3 TTS, Mayura:v1) + Web Speech API fallback for Odia & Hindi.

OCR & Vision

Gemini 2.5 Flash multimodal document OCR + OpenCV image pre-processing for CBC slips and ECG waveforms.

AI Intelligence Layer

Gemini 2.5 Flash for timeline extraction, structured summaries, and missing-info follow-up questions.

Safety & Governance

Deterministic rule engine (Python/JS), AES encryption, immutable clinician audit log, DPDP Act compliance.

How Safety is Engineered (Zero-Hallucination Mandate):

- Deterministic Urgency Rules: Deterministic rules set urgency; the LLM is summary-only with constrained Pydantic schemas. AI cannot downgrade an emergency (MAX(Rule, AI)).
- Dual-Source Verification: OCR and translation show confidence indicators alongside original verbatim text for instant human verification.
- Auditability & Explainability: Every flag stores its exact trigger rule (e.g. SpO2 < 90% -> RED), ensuring every decision is clinically explainable and auditable.

Prototype Architecture

End-to-end 4-stage dataflow with unified governance across every operational layer.

1. CAPTURE

Text / voice / report photo with explicit consent

- Patient / worker text
- Voice in Odia / Hindi
- Report photo / PDF
- Consent gate & 2D Body Map

2. PROCESS

Convert everything to clean, anonymised text

- Sarvam Speech-to-text
- Indic <> English translation
- OCR + document clean-up
- PII Anonymiser & ID token

3. UNDERSTAND + FLAG

Rules decide urgency; LLM only summarises

- Clinical entity extractor
- Chronological timeline builder
- Missing-info detector
- Deterministic Red Flag rules

4. HUMAN REVIEW

Qualified clinical staff decide, edit or escalate

- 3-lane priority queue
- Reviewer doctor dashboard
- ABDM referral note draft
- Emergency siren escalation

DATA GOVERNANCE & PRIVACY (Enforced Across All Layers):

Role-Based Access Control (Nurse Intake vs. Doctor Review) • Local AES-256 Encryption & Ephemeral Processing • Minimal Data Retention with Auto-Delete Policies • Deterministic Emergency Flag Skips
Routine Waiting Queues • Targeted Follow-up Questions Loop Directly Back to Frontline Worker

Execution Feasibility

High feasibility across technical, operational, and economic pillars with robust risk mitigation.

Technical Feasibility

Built on mature open-source tools & sovereign APIs; zero custom model training required; lightweight and easily maintained.

Operational Feasibility

Assists the frontline worker, never replaces the doctor; intuitive ergonomics require under 15 minutes of training.

Economic Feasibility

Open-source architecture; runs on modest local laptops or cloud tiers; negligible per-facility operating expense.

Key Risks & Built-in Architectural Mitigations:

- OCR errors on poor report images: Confidence indicators shown alongside raw photo; mandatory nurse verification before saving values.
- Translation nuances / dialect errors: Original verbatim regional audio and phonetic transliteration displayed beside translation for full clinical context.
- LLM hallucination on emergency triage: Summary-only role for generative AI; deterministic clinical rules alone compute and enforce urgency scores ($\text{MAX}(\text{Rule}, \text{AI})$).
- Low rural network connectivity: Offline-tolerant PWA with local IndexedDB queue and mirror rule engine; auto-syncs when connectivity returns.

 **Prototype Scope: Synthetic patient cohort and sample reports; includes multimodal intake, OCR, rules flags, and reviewer dashboard.**

Future Scalability

A 3-phase rollout roadmap with institutional scale-up levers for national public health impact.

Phase 1: Working Prototype

- Single facility interactive demo
- Multimodal intake (Voice + OCR + Body Map)
- Deterministic rules + AI summary
- Reviewer dashboard with synthetic cohort

Phase 2: Facility Clinical Pilot

- Deployment in 1 designated PHC / CHC
- Real clinical workflows & nurse shadowing
- Patient informed consent validation
- Medical officer usability feedback

Phase 3: District & State Scale

- Multi-facility unified dashboard
- Shared state clinical rule library
- Direct 108 ambulance referral network
- Connection to SCB Medical College



Institutional Scale-Up Levers:

- Multilingual Vernacular Expansion: Expanding to all 22 official Indian languages using sovereign models from Sarvam AI and Bhashini.
- On-Device Edge Models: Quantized lightweight SLMs (e.g. Gemma 2B via WebGPU) for 100% disconnected remote tribal health camps.
- ABDM / ABHA & HMIS Integration: Full M3 Milestone sync with National Health Authority health lockers and state e-Hospital portals.
- Setting-Specific Rule Packs: Specialized clinical rule packs for Maternal Care, Occupational Health, and Fever Outbreak Surveillance.
- Flexible Deployment Architecture: Cloud or on-premises deployment adhering to state government IT and DPDP Act privacy guidelines.

THANK YOU

Team: CODEX | Team ID: BH26PS07T060

BPUT Hackathon 2026: Multimodal Healthcare Triage Assistant for Government and Institutional Health Facilities

“Our goal: Help frontline healthcare staff prioritize patients faster and safer, while keeping clinical decisions strictly in human hands.”

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